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Quick lock release coupling

Abstract

A quick release coupling for providing a connection between a pair of hydraulic lines is provided. The coupling includes a male component has a fluid passage therethrough and includes a shank portion adapted to engage a first hydraulic line. The shank portion has a first port for establishing fluid communication between the first hydraulic line and the fluid passage. The shank portion also has a restraining mechanism adapted to engage the first hydraulic line and block a rotational movement and an axial movement of the male component relative to the first hydraulic line. The coupling also includes a coupling adapter connectable to the male component and having a fluid channel therethrough adapted to be in fluid communication with the fluid passage. The coupling adapter is adapted to engage a second hydraulic line and has a second port for establishing fluid communication between the second hydraulic line and the fluid channel.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a National Stage of International patent application PCT/CA2022/050686, filed on May 3, 2022, which claims priority to U.S. Provisional Patent Application Nos. 63/201,573 and 63/201,574 filed May 5, 2021, the disclosures of which are incorporated by reference in their entireties.

TECHNICAL FIELD

(1) The present disclosure generally relates to couplings, and more particularly, to quick lock release couplings for use with hydraulic lines.

BACKGROUND

(2) Several types of couplings are used in the fluid transfer industry. For example, push-to-connect connections, or “quick-release couplings”, are different even though the applications and some of the structure are similar to other known couplings. Widely used in the pneumatic industry and for low-pressure applications, quick-release couplings are not designed for higher-pressure applications, such as for the transfer of fluids on heavy machinery where pressurized fluid are flowing and where torsion constraints are present. Hydraulic lines, for instance, being for the most part constructed with braded or intertwined metallic wires or synthetic plastic material, are quite sensitive to torsion constraints.

(3) Swivelling couplings are components that enable two or more hydraulic lines to be interconnected axially or angularly, and to reduce torsion constraints that could affect their physical integrity or internal structure. Quick-release couplings often consist of a male connector with a groove around its periphery allowing the ball bearings of the female connector to be trapped in said groove. Thus, the axial movement is eliminated, while the assembly is free to rotate. Therefore, it becomes similar in every way to a rotary joint consisting of one or more rows of balls and a seal. Pressurized hydraulic fluid, along with the rotation of known rotary joints and swivelling

couplings, accelerates wear and tear of the various seals within the coupling, which requires frequent maintenance, repair and replacement.

SUMMARY

(4) According to an aspect, a quick release coupling for providing a connection between a first fluid line provided with a female component comprising a ball bearing assembly and a second fluid line is provided. The quick release coupling includes a male component having a fluid passage therethrough and comprising a body portion having an internal surface at least partially defining the fluid passage; and a shank portion extending from the body portion and being adapted to engage the female component of the first fluid line, the shank portion having a shank portion port adapted to establish fluid communication between the first fluid line and the fluid passage, the shank portion comprising a restraining mechanism adapted to engage the ball bearing assembly and block a rotational movement and an axial movement of the male component relative to the first fluid line when engaging the first fluid line. The quick release coupling also includes a coupling adapter having a fluid channel therethrough and being securable to the male component proximate the body portion to establish fluid communication between the fluid channel and the fluid passage, the coupling adapter being adapted to engage the second fluid line and comprising an adapter port adapted to establish fluid communication between the second fluid line and the fluid channel.

(5) According to a possible embodiment, the shank portion comprises an external surface, and wherein the restraining mechanism is provided along the external surface of the shank portion.

(6) According to a possible embodiment, the restraining mechanism comprises a groove extending circumferentially about the shank portion, the groove being shaped and adapted to receive the ball bearing assembly for blocking the axial movement of the male component.

(7) According to a possible embodiment, the groove extends circumferentially about the shank portion in a single plane.

(8) According to a possible embodiment, the groove extends partially about the shank portion such that the groove comprises a first end and a second end spaced from one another and defining an ungrooved portion therebetween.

(9) According to a possible embodiment, the ungrooved portion is shaped and adapted to engage the ball bearing assembly between a pair of adjacent bearing balls to block the rotational movement of the male component.

(10) According to a possible embodiment, the restraining mechanism further comprises a detent adapted to engage the ball bearing assembly to block the rotational movement of the male component.

(11) According to a possible embodiment, the detent comprises a plurality of recesses provided along the groove for receiving respective bearing balls of the ball bearing assembly to block the rotational movement of the male component.

(12) According to a possible embodiment, the detent comprises a protrusion extending from the shank portion, the protrusion being shaped and sized to engage the ball bearing assembly between a pair of adjacent bearing balls to block the rotational movement of the male component.

(13) According to a possible embodiment, the protrusion extends from within the groove.

(14) According to a possible embodiment, the protrusion is integrally formed with the shank portion or is welded thereto.

(15) According to a possible embodiment, the quick release coupling further includes a valve assembly operable to control fluid flow along the fluid passage of the male component before, during and after engaging the shank portion with the first fluid line.

(16) According to a possible embodiment, the valve assembly comprises a valve head provided within the fluid passage proximate the shank portion port and being operable in a closed position where fluid flow is blocked through the shank portion port, and an open position, the valve assembly further comprising a head spring connected to the valve head and being adapted to bias the valve head in the closed position.

- (17) According to a possible embodiment, the valve assembly comprises a valve body provided within the fluid passage along the body portion and being operable in a closed position where fluid flow through the fluid passage is blocked, and an open position where fluid flow through the fluid passage is allowed, the valve assembly further comprising a body spring connected to the valve body and being adapted to bias the valve body in the closed position.
- (18) According to a possible embodiment, the coupling adapter further comprises a housing having internal surfaces comprising at least one radial surface and at least one axial surface, said internal surfaces defining a cavity having an open end, wherein the body portion of the male component is shaped and sized to engage the cavity via the open end and abut the at least one radial surface.
- (19) According to a possible embodiment, the quick release coupling further includes a nut securable within the cavity of the housing and surrounding a section of the shank portion to radially restrain the male component, the nut being adapted to axially restrain the body portion within the cavity and allow the male component to rotate with respect to the housing and the nut about a longitudinal axis of the shank portion, thereby enabling the first and second fluid lines to be swivellably interconnected.
- (20) According to a possible embodiment, the quick release coupling further includes a thrust washer surrounding the shank portion between the nut and the body portion.
- (21) According to a possible embodiment, the shank portion comprises a flange extending radially outward and having a flange surface being substantially perpendicular to the external surface of the shank portion and facing the nut, and wherein the quick release coupling further comprises an outer sealing element surrounding the shank portion and extending between the flange surface and the nut.
- (22) According to another aspect, a male component of a quick release coupling for connection with a hydraulic line provided with a female component comprising a ball bearing assembly is provided, with the male component having a fluid passage therethrough and comprising a shank portion extending along a longitudinal axis and being adapted to engage the hydraulic line, the shank portion having a shank portion port adapted to establish fluid communication between the hydraulic line and the fluid passage, the shank portion comprising a restraining mechanism adapted to engage the ball bearing assembly to block a rotation of the male component about the longitudinal axis.
- (23) According to a possible embodiment, the shank portion comprises an external surface, and wherein the restraining mechanism is provided along the external surface of the shank portion.
- (24) According to a possible embodiment, the restraining mechanism comprises a groove extending circumferentially about the shank portion, the groove being shaped and adapted to receive the ball bearing assembly for blocking an axial movement of the male component along the longitudinal axis.
- (25) According to a possible embodiment, the restraining mechanism further comprises a detent adapted to engage the ball bearing assembly to block the rotational movement of the male component.
- (26) According to a possible embodiment, the detent comprises a plurality of recesses provided along the groove for receiving respective bearing balls of the ball bearing assembly to block the rotational movement of the male component.
- (27) According to a possible embodiment, the detent comprises a protrusion extending from the shank portion, the protrusion being shaped and sized to engage the ball bearing assembly between a pair of adjacent bearing balls to block the rotational movement of the male component.
- (28) According to a possible embodiment, the protrusion extends from within the groove.
- (29) According to a possible embodiment, the protrusion is integrally formed with the shank portion or is welded thereto.
- (30) According to another aspect, a quick release coupling for providing a connection between a pair of hydraulic lines is provided. The quick release coupling includes a male component having a

fluid passage therethrough and comprising a shank portion adapted to engage a first hydraulic line, the shank portion having a first port adapted to establish fluid communication between the first hydraulic line and the fluid passage, the shank portion comprising a restraining mechanism adapted to cooperatively engage the first hydraulic line and block a rotational movement and an axial movement of the male component relative to the first hydraulic line when engaged with the first hydraulic line. The quick release coupling also has a coupling adapter connectable to the male component and having a fluid channel therethrough adapted to be in fluid communication with the fluid passage, the coupling adapter being adapted to engage a second hydraulic line and comprising a second port adapted to establish fluid communication between the second hydraulic line and the fluid channel.

(31) According to another aspect, a use of a quick release coupling for providing a connection between a pair of hydraulic lines is provided. The quick release coupling includes a male component having a fluid passage therethrough and comprising a shank portion adapted to engage a first hydraulic line, the shank portion having a first port adapted to establish fluid communication between the first hydraulic line and the fluid passage, the shank portion comprising a restraining mechanism adapted to cooperatively engage the first hydraulic line and block a rotational movement and an axial movement of the male component relative to the first hydraulic line when engaged with the first hydraulic line. The quick release coupling also includes a coupling adapter connectable to the male component and having a fluid channel therethrough adapted to be in fluid communication with the fluid passage, the coupling adapter being adapted to engage a second hydraulic line and comprising a second port adapted to establish fluid communication between the second hydraulic line and the fluid channel.

(32) According to a possible embodiment, the quick release coupling is adapted to operate between about 0 psi and 5000 psi.

(33) According to another aspect, a use of the quick release coupling defined above for providing a connection between a pair of hydraulic lines is provided.

(34) According to another aspect, a method of connecting a first fluid line to a second fluid line using a quick release coupling as defined above is provided. The method includes connecting the coupling adapter to the second fluid line; and engaging the shank portion with the first fluid line to block a rotational movement and an axial movement of the quick release coupling relative to the first fluid line.

(35) According to another aspect, a method of connecting a first fluid line provided with a female component to a second fluid line is provided. The method includes connecting a first end of a hydraulic coupling to the second fluid line; and connecting a second end of the hydraulic coupling to the first fluid line, the second end being provided with a male component configured to engage the female component and block a rotational movement and an axial movement of the male component relative to the female component.

(36) According to a possible embodiment, the hydraulic coupling is a quick release coupling.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a quick lock release coupling provided with a restraining mechanism for preventing movement of a portion of the quick lock release coupling, according to a possible embodiment.

(2) FIG. 2 is a side view of the quick lock release coupling shown in FIG. 1, showing a male component and a coupling adapter extending at opposite ends of the quick lock release coupling, according to an embodiment.

(3) FIG. 3 is a front view of the quick lock release coupling shown in FIG. 1, showing a valve head

provided within a port of the coupling, according to an embodiment.

(4) FIG. 4 is a sectional view of the quick lock release coupling shown in FIG. 3, taken along line 4-4, showing a fluid passage extending through a housing of the quick lock release coupling, according to an embodiment.

(5) FIGS. 5 to 8 are top views of various embodiments of a male component, showing possible embodiments of the restraining mechanism.

(6) FIG. 9 is a perspective view of a quick lock release coupling, according to an alternate embodiment.

(7) FIG. 10 is a side view of the quick lock release coupling shown in FIG. 9, showing a male component and a coupling adapter extending at opposite ends of the quick lock release coupling, according to an embodiment.

(8) FIG. 11 is a front view of the quick lock release coupling shown in FIG. 9, showing a valve head provided within a port of the coupling, according to an embodiment.

(9) FIG. 12 is a sectional view of the quick lock release coupling shown in FIG. 11, taken along line 12-12, showing a pressure eliminator provided within a valve body, according to an embodiment.

(10) FIG. 13 is a perspective view of a swivelling coupling, according to an alternate embodiment.

(11) FIG. 14 is a side view of the swivelling coupling shown in FIG. 13, showing the male component coupled within the coupling adapter via a nut, according to an embodiment.

(12) FIG. 15 is a front view of the swivelling coupling shown in FIG. 13, showing a valve head provided within the port of the male component, according to an embodiment.

(13) FIG. 16 is a sectional view of the swivelling coupling shown in FIG. 15, taken along line 16-16, showing a flange portion of the male component provided between internal surfaces of the coupling adapter housing and the nut, according to an embodiment.

(14) FIG. 17 is a perspective view of a swivelling coupling, according to an alternate embodiment.

(15) FIG. 18 is a side view of the swivelling coupling shown in FIG. 17, showing the male component coupled within the coupling adapter via a nut, according to an embodiment.

(16) FIG. 19 is a front view of the swivelling coupling shown in FIG. 17, showing a male component port, according to an embodiment.

(17) FIG. 20 is a sectional view of the swivelling coupling shown in FIG. 19, taken along line 20-20, showing the shank portion of the male component being coupled to the flange portion provided within the coupling adapter housing, according to an embodiment.

(18) FIG. 20A is an enlarged view of a portion of the swivelling coupling shown in FIG. 20, showing a pair of sealing rings provided about the shank portion between the shank portion and the nut, according to an embodiment.

(19) FIG. 21 is a perspective view of a swivelling coupling, according to an alternate embodiment.

(20) FIG. 22 is a sectional view of the swivelling coupling shown in FIG. 21, showing a shank portion defining a female connection, according to an embodiment.

(21) FIGS. 23 to 25 are top views of couplings, showing a coupling cartridge (FIG. 23), a swivelling coupling (FIG. 24) and a quick release male coupling (FIG. 25) according to possible embodiments.

(22) FIG. 25B is a perspective view of a multiport swivel assembly provided with various implementations of couplings, according to an embodiment.

(23) FIG. 25C is a sectional view of an alternative embodiment of a coupling, showing a threaded connection port.

(24) FIG. 26 is a top view of a pair of swivelling couplings coupled to a multiport swivel assembly, according to an embodiment.

(25) FIG. 27 is a perspective view of a coupling provided with a crimp fitting, according to an alternate embodiment.

(26) FIG. 28 is a side view of the coupling shown in FIG. 27, showing the coupling adapted and the

crimp fitting extending at opposite ends of the coupling, according to an embodiment.

(27) FIG. 29 is a front view of the coupling shown in FIG. 27, according to an embodiment.

(28) FIG. 30 is a sectional view of the quick lock release coupling shown in FIG. 29, taken along line 30-30, showing a fluid passage extending through a housing of the quick lock release coupling, according to an embodiment.

(29) FIGS. 31 and 32 are representations of a possible application comprising one or more couplings, showing a fishing vessel pump prior to being submerged (FIG. 31) and the fishing vessel pump underwater for operation (FIG. 32).

(30) FIG. 33 is an illustration of a hydraulic machine comprising couplings installed in a vertical configuration, according to a possible embodiment.

(31) FIGS. 34 and 35 are perspective views of an alternate embodiment of a coupling, showing latching elements (FIG. 35) having an elongate shape and extending partially circumferentially within ports of a male component (FIG. 34).

DETAILED DESCRIPTION

(32) As will be explained below in relation to various implementations, the present disclosure describes devices and systems for providing improved connections for hydraulic machinery, such as quick lock release couplings configured to be coupled to and between hydraulic lines. The present disclosure describes a quick lock release coupling that is easily and reliably connectable to hydraulic lines adapted for the transport of hydraulic fluid at elevated pressures. The quick lock release coupling can include a male component shaped and sized to engage a female component of a hydraulic line. The male component includes a restraining mechanism adapted to block movement of the male component relative to the hydraulic line (e.g., relative to the female component) when coupled thereto. The female component traditionally includes a ball bearing assembly adapted to prevent axial movement of the male component while enabling rotation of the male component when coupled thereto. The restraining mechanism described herein can be further adapted to block a rotational movement of the male component. For example, the restraining mechanism can include a detent shaped and sized to extend between a pair of adjacent balls of the ball bearing assembly, thereby preventing rotation of the male component.

(33) It is thus noted that the male component is adapted to remain generally static relative to the hydraulic line due to the restraining mechanism. Moreover, it is appreciated that blocking movement of the male component relative to the hydraulic line can improve the lifespan and efficiency of the various sealing components of the quick lock release coupling, which sustain less stress, less friction, less compression forces, etc., due to the static assembly of the male component.

(34) The quick lock release couplings can be configured to enable the transport of hydraulic fluid for various operations. The couplings can be implemented on various apparatuses, machines and devices, such as forestry combo harvesters, industrial reach mowers, compact tracked loader attachments, agriculture attachments, among others. As will be described herein, the hydraulic coupling can be adapted to provide interconnection between two fluid lines. The coupling can enable a coaxial interconnection of two fluid lines, but it should be understood that the coupling may be adapted to provide a different interconnection, such as a 90° connection or an oblique connection. Alternatively, or additionally, the quick lock release coupling can include an integrated swivel assembly adapted to enable motion of the one or more hydraulic lines connected to the quick lock release coupling and provide long-lasting sealing capabilities before servicing.

(35) With reference to FIGS. 1 to 4, an example quick release coupling 10 (or simply “coupling”) for interconnecting and establishing fluid communication between a pair of fluid lines is illustrated. The coupling 10 includes one or more components connected to one another and defining a central passage 12 through the coupling 10. Therefore, fluid, such as hydraulic fluid, can flow through the coupling 10 via the central passage 12. In some embodiments, the coupling 10 includes a first component configured to engage a first fluid line, and a second component configured to engage a second fluid line, thereby connecting the fluid lines to one another. The first and second

components are secured to one another and each define a portion of the central passage **12**. In this embodiment, the central passage **12** is generally longitudinal, with the inlet of the first component being substantially aligned with the inlet of the second component. However, it is appreciated that other configurations are possible, such as having a curved or bent central passage **12**.

(36) In this embodiment, the first component of the coupling **10** can include a male component **14** defining a fluid passage **15** therethrough, and having a body portion **16** and a shank portion **18** extending from a first side of the body portion **16**. As seen in FIG. **4**, the male component **14** has internal surfaces **20** extending along the body and shank portions **16**, **18** which define the fluid passage **15**. Furthermore, and as will be described further below, the shank portion **18** is adapted to engage the first fluid line and includes a first port, or male component port **22**, configured to establish fluid communication between the first fluid line and the fluid passage **15**.

(37) In this embodiment, the second component of the coupling **10** includes a coupling adapter **24** defining a fluid channel **25** therethrough. The coupling adapter **24** is securable to the male component **14** in a manner such that the fluid channel **25** communicates with the fluid passage **15**. It is noted that, in this embodiment, the combination of the fluid passage **15** and the fluid channel **25** forms the central passage **12** of the coupling **10**, although other configurations are possible. In this embodiment, the coupling adapter **24** is adapted to be secured to the body portion **16** of the male component **14**, opposite the shank portion **18**. For example, the body portion **18** and the coupling adapter **24** can have complementarily shaped threads, enabling the coupling adapter **24** to be threaded into the body portion **18**. In some embodiments, the coupling **10** includes an adapter seal **28** provided between the coupling adapter **24** and male component **14** to further secure the components together and prevent fluid from leaking therebetween. As seen in FIGS. **2** and **4**, the coupling adapter **24** is provided with an adapter port **26** adapted to engage the second fluid line and establish fluid communication between the second fluid line and the fluid channel **25**. It is thus noted that the first fluid line and the second fluid line are fluidly connected via the coupling **10**, where fluid flows through the fluid channel **25** and fluid passage **15**, in either direction.

(38) Still referring to FIGS. **1** to **4**, the coupling **10** can be provided with a valve assembly **30** configured to control the flow of fluid through the coupling **10** (e.g., through the male component **14**, through the coupling adapter **24** or a combination thereof). The valve assembly **30** can be operable between a closed configuration, where fluid flow through the first port **22** is blocked, and an open configuration, wherein fluid flow through the first port **22** is allowed. In this embodiment, the valve assembly **30** includes a valve head **32** provided along the fluid passage **15**, and more specifically within the first port **22**. The valve head **32** is adapted to move between a closed position (seen in FIG. **4**) where the valve head **32** obstructs the first port **22**, and an open position, where the valve head **32** is displaced such that the first port **22** becomes unobstructed.

(39) In this embodiment, the valve assembly **30** includes a resilient element **34** coupled to the valve head **32** and configured to bias the valve head **32** in the closed position. For example, the resilient element **34** can include a spring **35** operatively connected to the valve head **32** to bias the valve head **32** in the closed position. As seen in FIG. **4**, the spring **35** can extend along the fluid passage **15** and abut against an internal surface of the coupling adapter **24**. It is noted that opening the valve assembly **30** (i.e., moving the valve head **32** in the open position) includes moving the valve head **32** further within the fluid passage **15** (i.e., towards the coupling adapter **24**), thereby compressing the spring **35** and opening the first port **22**. In some embodiments, the female component can be provided with an actuator, such as a stem, configured to push against the valve head **32** when engaging the shank portion **18** with the first fluid line. In other words, while being pressed against the stem of the female component, the valve head retreats inside the male component, thereby opening the first port **22**. The valve head **32** illustratively has a flat outer surface **36** adapted to be generally coplanar with the end of the shank portion **18** when in the closed position.

(40) In some embodiments, the shank portion **18** includes a restraining mechanism **40** adapted to prevent undesired or accidental disengagement of the male component **14** from the first fluid line.

For example, the restraining mechanism **40** can be provided along the external surface of the shank portion **18** for engaging an inner surface of the first fluid line. Traditionally, the first fluid line can include a female component (not shown) provided with a ball bearing assembly installed circumferentially along the inner surface. Moreover, and as seen in FIGS. **1**, **2** and **4**, the restraining mechanism **40** includes a groove **42** extending circumferentially about the shank portion **18**, and being shaped and sized to receive the ball bearing assembly therein. As such, once the shank portion **18** engages the first fluid line, the ball bearing assembly and the groove **42** cooperate with one another to block axial movement of the male component **14** relative to the first fluid line, thereby preventing disengagement therefrom.

(41) In some embodiments, the restraining mechanism **40** is further adapted to block a rotational movement of the male component **14** relative to the first fluid line. More particularly, it is noted that the ball bearing assembly can enable rotation of the male component **14** about its longitudinal axis (A). However, the restraining mechanism **40** can include a detent **44** shaped and sized to engage the ball bearing assembly to prevent rotation of the male component **14**. The detent **44** can be fixedly connected to the shank portion **18** such that engagement of the detent **44** with the ball bearing assembly prevents rotation of the shank portion **18** (and thus of the male component **14**) in either direction about the longitudinal axis. As seen in FIG. **2**, the detent **44** can include a protrusion **46** extending from the shank portion **18** and which is shaped and sized to extend between a pair of adjacent ball bearings to block the rotation of the male component relative to the first fluid line.

(42) In this embodiment, the protrusion **46** extends from within the groove **42** to facilitate its engagement with the ball bearing assembly (e.g., between a pair of adjacent ball bearings). Furthermore, the protrusion **46** can be integrally formed as part of the shank portion **18** (e.g., during machining of the groove **42**), although it is appreciated that other configurations are possible. For example, and with reference to FIGS. **5** to **8**, the detent **44** can be provided about the shank portion **18** between the body portion **16** and the groove **42** (see FIGS. **5** and **6**), or within the groove **42** (see FIGS. **7** and **8**). In the embodiment of FIG. **5**, the detent **44** can include a ball **48**, or dome-shaped element, connected to and extending from the shank portion **18** to engage the female component of the first fluid line. It should be noted that the female component may require an additional complementarily shaped element configured to engage with the ball **48** to block the rotational movement of the male component. The ball **48** can be spot welded onto the shank portion **18**, integrally formed therewith or connected thereto using any suitable method.

(43) Alternatively, and with reference to FIG. **6**, the detent **44** can include a protuberance **50** provided between the body portion **16** and the groove **42** along the external surface of the shank portion **18**. In a similar fashion as the ball **48**, the protuberance **50** is configured to engage a complementarily-shaped element of the female component for blocking the rotation of the male component. In some embodiments, the complementarily-shaped element includes a recess (not shown) having a shape conforming to the shape of the protuberance **50** such that the male component **14** is adapted to engage the female component in a predetermined orientation. It is appreciated that the male component **14** can include a plurality of detents **44** provided about the shank portion **18**, both in and/or proximate the groove **42**.

(44) In yet another embodiment, and as seen in FIG. **7**, the detent **44** can include one or more recesses **52** provided about the shank portion **18** and/or within the groove **42**. The recesses **52** are shaped and sized to receive a ball bearing of the ball bearing assembly such that rotation of the male component **14** about the longitudinal axis is blocked. It is noted that the ball bearing assembly includes a finite number of ball bearings, and that the detent **44** can include any suitable number of recesses **52**, such as fewer than the number of ball bearings, more than the number of ball bearings, or the same number. For example, the ball bearing assembly can include twelve (12) ball bearings, and the detent **44** can include twenty-four (24) recesses **52** to facilitate engagement of the ball bearings with the detent.

(45) Now referring more specifically to FIG. 8, in this embodiment, the detent **44** includes the protrusion **46** provided along the groove **42**, and further includes the ball **48** connected to and extending from the protrusion **46**. The protrusion **46** can be adapted to prevent rotation of the male component **14** when engaged with the female component, while the ball **48** can facilitate positioning the male component **14** to have a pair of adjacent ball bearings on either side of the protrusion **46**. It should be noted that the detent **44** of the restraining mechanism **40** can include any suitable component, feature or combination thereof to prevent rotation of the male component **14**. For example, the detent **44** can include the recesses **52** provided along the groove **42**, in addition to having a protuberance **50** extending from the body portion **18** and between a pair of adjacent ball bearings. It is appreciated that any other combination of the detent **44** can be used and is possible.

(46) Now referring to FIGS. 9 to 12, another embodiment of the coupling **10** will be described. The shank portion **18**, and related components (e.g., the restraining mechanism **40**) can be substantially the same as the previously described embodiment, and adapted to engage the first fluid line, and prevent axial and rotational movement of the male component. In this embodiment, the body portion **16** is elongated and includes internal surfaces adapted to define one or more internal cavities **60** along the fluid passage **15**. Referring more specifically to FIG. 12, the internal cavities **60** can include a first cavity **62** proximate the shank portion **18**, and a second cavity **64** proximate the coupling adapter **24** when the coupling adapter **24** is connected to the male component **14**.

(47) In this embodiment, the valve assembly **30** further includes a valve body **70** installed along the fluid passage **15**, such as within the second cavity **64**, and being adapted to block fluid flow between the first and second cavities **62**, **64**. The valve body **70** is adapted to move between a closed position where the valve body **70** obstructs the fluid passage **15** between the first and second cavities **62**, **64**, and an open position, where the valve body **70** is displaced such that fluid flow is enabled. As seen in FIG. 12, the internal surfaces of the body portion **16** can include abutment surfaces **17** shaped and sized to have the valve body **70** abut thereon for blocking fluid flow through the fluid passage **15**. The valve assembly **30** can also include a second resilient element **72**, such as a second spring **73**, configured to bias the valve body **70** in the closed position (e.g., bias the valve body **70** against the abutment surfaces **17**).

(48) In addition, the valve assembly **30** can include a valve body actuator **71** configured to displace the valve body **70** from the closed position to the open position, thereby compressing the second spring **73** and enabling fluid flow. In this embodiment, the valve body actuator **71** includes a shaft **74** connected to and extending from the valve head **32** within the fluid passage **15**. It is thus noted that engaging the male component with the female component of the first fluid line causes the valve head **32** to retreat within the fluid passage **15**, and subsequently causes the shaft **74** to engage and push against the valve body **70** to move it in the open position. Therefore, the shank portion **18** can at least partially engage the female component prior to fluid communication being established between the first and second fluid lines due to the distance between the shaft head **77** and the valve body **70**.

(49) In some embodiments, it can be desirable to at least partially control the pressure within the coupling **10**, such as within the fluid channel **25** and the fluid passage **15** (e.g., within the first cavity **62** and/or the second cavity **64**) to prevent malfunctions or damage caused by pressure differentials or pressurization issues. For example, hydraulic fluid can be provided to the second cavity **64** via the second fluid line, with the valve body **70** blocking fluid flow into the first cavity **62**. Therefore, it is noted that the pressure within the second cavity **64** increases, while the pressure within the first cavity **62** remains low (e.g., null), making it difficult to move the valve body **70** in the open position to provide hydraulic fluid to the first fluid line. As such, in some embodiments, the valve body **70** can include a pressure eliminator **76** operable to eliminate, or at least reduce, the pressure within the first and second cavities **62**, **64** prior to moving the valve body **70** in the open position.

(50) In this embodiment, and still referring to FIG. 12, the valve body **70** has a bore **75** defined

therethrough and communicating with the first cavity **62** at a first end thereof, and with the second cavity **64** at a second end thereof. The pressure eliminator **76** can be coupled to the valve body **70** within the bore **75** and is operable to selectively eliminate the pressure within the second cavity to facilitate opening the valve body **70** (e.g., moving to the open position). More specifically, the pressure eliminator **76** can include an obstructing component, such as a plug **78**, provided within the bore **75** and being shaped and configured to block fluid flow therethrough. The plug **78** can be displaced to enable a restricted flow of fluid from the second cavity **64** to the first cavity **62**, thereby reducing (e.g., eliminating) the pressure within the second cavity **64**, or at least creating a pressure balance between the first and second cavities (e.g., substantially the same pressure in each cavity). Therefore, displacement of the valve body **70** into the second cavity **64** to operate it in the open position is facilitated.

(51) In some embodiments, the bore **75** can include a seat **80** adapted to have the plug **78** sit thereon when obstructing fluid flow through the bore **75**. As seen in FIG. **12**, in this embodiment, the seat **80** is provided proximate the first end of the bore **75** (i.e., proximate the first cavity **62**), and the plug **78** engages the seat **80** to block fluid flow through the bore **75**. It is noted that the plug **78** is movable between a seated position, where the plug **78** engages the seat **80** to block fluid flow, and an unseated position, where the plug **78** is displaced and spaced from the seat **80**, thereby enabling fluid flow between the first and second cavities **62**, **64**. In this embodiment, the pressure eliminator **76** further includes a plug biasing element **82** operatively connected to the plug **78** and being configured to bias the plug **78** in the seated position. As will be described further below, the pressure eliminator **76** can be operated to apply a force on the plug **78** in order to displace and unseat it. Once the force is removed, the plug biasing element **82** is configured to revert the plug **78** in the seated position, thereby blocking fluid flow through the bore **75** once more.

(52) In this embodiment, the pressure eliminator **76** further includes a cap **84** shaped and sized to fit within the bore **75** to at least partially restrict fluid flow therethrough. In this embodiment, the cap **84** is connected to the valve body **70** within the bore **75** via interference fit, although other connection methods are possible, such as via fasteners, via a keyjoint connection, via adhesive, etc. The cap **84** is illustratively provided proximate the second end of the bore **75**, and is thereby on the side of the second cavity **64**. As seen in FIG. **12**, the plug biasing element **82** is connected to the cap **84** and extends between the cap **84** and the plug **78**. Furthermore, the cap **84** includes a cap passage **85** defined therethrough for enabling fluid flow into the bore **75** from the second cavity **64**. Therefore, when fluid is introduced within the second cavity **64** (e.g., via the second fluid line), the fluid can flow into the bore **75** via the cap passage **85**, although fluid flow into the first cavity **62** is blocked by the plug **78** when in the seated position.

(53) As previously described, engaging the coupling **10** with the female component of the first fluid line causes the valve head **32** to retreat within the fluid passage **15**, and subsequently causes the shaft head **77** to engage and push against the valve body **70**. Moreover, when in the seated position, the plug **78** can have a portion thereof extend into the first cavity **62**. For example, in the illustrated embodiment of FIG. **12**, the plug **78** includes a generally spherical body **79** configured to engage the seat **80** in a manner such that a portion of the spherical body **79** communicates with the first cavity **62**. As such, moving the shaft **74** toward the valve body **70** causes the shaft head **77** to engage and push the plug **78** inwardly within the bore **75** (e.g., toward the second cavity **64**) prior to engaging the valve body **70**. It is thus noted that a restricted flowrate of fluid is enabled between the first and second cavities **62**, **64** through the bore **75** due to the unseating of the plug **78** via the shaft head **77**. It is further noted that enabling the fluid flow between the cavities **62**, **64** causes the pressure within the second cavity **64** to drop, thereby facilitating displacement of the valve body **70** into the second cavity **64**, and into the open position, therefore establishing a greater flowrate of fluid into the first cavity, and thus between the first and second fluid lines.

(54) It is appreciated that the coupling **10** includes seals **90**, or sealing elements, provided between various components for preventing fluid flow through the coupling **10** when fluid flow is undesired

(e.g., when the coupling **10** is connected to only one of the first and second fluid lines). For example, the coupling **10** can include a shank portion seal **91** provided proximate the male component port **22** such that the shank portion seal **91** engages the valve head **32** prior to engaging the coupling **10** with the first fluid line, and as seen in FIG. **12**. In addition, the coupling **10** can include an internal seal **92** provided about the valve body **70** such that when the valve body **70** is in the closed position, the internal seal **92** engages the abutment surfaces **17** to prevent fluid flow through interstices between the valve body **70** and the body portion **16**. It is appreciated that the coupling **10** can include any suitable number of seals provided between any abutting components thereof.

(55) The above-described implementations of the coupling provide for a quick-connect coupling, or a cartridge therefor (seen in FIG. **17**), configured to connect to a complementary portion (e.g., male and/or female portion) and define a static connection therebetween. Particularly, the coupling **10** is adapted to block axial movement between the male and female portions, via the engagement of the ball bearings of the female portion with the groove of the male portion, for example. In addition, the restraining mechanism **40** is configured to block a rotational movement of the male and female components relative to one another. As such, it is appreciated that the coupling is adapted to block both axial and rotational movement between the male and female components, thereby defining the static connection therebetween. It should be noted that providing a static connection can increase lifespan of the coupling by reducing the stress, friction among other forces applied to various components of the coupling during operation.

(56) Now referring to FIGS. **13** to **16**, another implementation of the coupling is illustrated. In this embodiment, the coupling can be a swivelling coupling **100**, where the male component **14** is rotatably coupled to the coupling adapter **24** such that the first fluid line can rotate relative to the second fluid line. In other words, the coupling **100** provides a swivelling connection between the first and second fluid lines. In this embodiment, the male component **14** includes a flange portion **102** extending radially outward from the body portion **16**, at an end thereof opposite the shank portion **18**. Furthermore, in this embodiment, the coupling adapter **24** includes a housing **104** defining an adapter cavity **105** shaped and adapted to receive the body portion **16** of the male component **14** therein. As will be further described below, the swivelling coupling **100** also includes a nut **110** insertable within the housing **105** for at least partially securing the male component **14** therein.

(57) In some embodiments, the adapter cavity **105** includes internal surfaces including at least one radial surface **106** and at least one axial surface **108**. It should be understood that, as used herein, the expression “radial surface(s)” can refer to surfaces extending in generally perpendicular planes relative to the longitudinal axis (A) of the coupling **100**. Similarly, it should be understood that, as used herein, the expression “axial surface(s)” can refer to surfaces being generally parallel to the longitudinal axis (A), and therefore perpendicular relative to the radial surfaces. The internal surfaces **106**, **108** define the adapter cavity **105** which has an open end **107** through which the male component **14** is insertable. However, it is appreciated that other configurations are possible, for example, the housing **104** may include a plurality of cavities in which a corresponding number of male components and nuts may be introduced. The internal surfaces **106**, **108** are preferably integrally formed with each other. This usually results from the machining of the housing **104** from a solid piece. In addition, the housing **104** includes an external surface **112** having any suitable finish and/or form. For example, the external surface **112** can have a generally cylindrical or circular shape. In the illustrated embodiment of FIGS. **13** to **16**, the external surface **112** is curved and adapted to be retained within a retaining ring (not shown) mounted on the hydraulic machine and enable rotational freedom relative to the retaining ring.

(58) In some embodiments, inserting the body portion **16** within the open end **107** of the adapter cavity **105** includes having the flange portion **102** cooperatively abut on the internal surfaces of the adapter cavity **105**. Still referring to FIG. **16**, it is appreciated that engaging the male component **14**

within the housing **104** can establish fluid communication between the fluid passage **15** of the male component **14** and the fluid channel **25** of the coupling adapter **24**. In this embodiment, the fluid passage **15** and fluid channel **25** are substantially collinear, although it is appreciated that other configurations are possible and may be used. For example, the fluid passage **15** and fluid channel **25** may have a variety of orientations, depending on the desired application, such as 90° angles or oblique angles.

(59) In this embodiment, the flange portion **102** is integrally formed with the body portion **16** and extends therefrom in a manner such that the male component **14** has a generally T-shape. The flange portion **102** can be radially continuous and symmetrical, and can be disk-shaped and extend perpendicularly with respect to the body portion **16**. As will be described below, this configuration of the male component **14** can provide support surfaces for abutting on the internal surfaces of the housing **104** and the nut **110**, thereby distributing the forces (i.e. decreasing the pressure) applied on the components of the coupling **100**. However, it is appreciated that the flange portion **102**, and/or any other component of the male component **14**, can have any suitable shape which cooperates with the internal surfaces of the housing **104**. In some embodiments, the flange portion **102** can include scores (not shown) defined on an outer circumferential surface thereof for enabling fluid to flow between the flange portion **102** and the housing **104**. This can be useful for lubricating purposes and for defining a self-lubricating coupling, such as the one described in Applicant's U.S. Pat. No. 8,047,579, which is incorporated herein by reference.

(60) Still referring to FIGS. **13** to **16**, the nut **110** is securable within the adapter cavity **105** of the housing **104** and surrounding the male component **14** to radially restrain the same within the adapter cavity **105**. In some embodiments, the nut **110** can be adapted to surround the male component **14** from the port **22** to the flange portion **102**, which allows for improved support, stability and resistance to forces. In this embodiment, the nut **110** is secured about the body portion **16** of the male component **14**, with the shank portion **18**, and thus the port **22**, extending beyond the nut **110** to facilitate connection with the first fluid line. In this embodiment, the nut **110** includes a projection **114** extending axially and internally into the cavity **105** of the housing **104** for axially restraining the flange portion **102**, while allowing the male component **14** to rotate with respect to the housing **104** and the nut **110**. The rotation of the male component **14** occurs about a longitudinal axis (A) of the shank portion **18**.

(61) In some embodiments, the nut **110** can have external threads and the internal surfaces of the housing **104** (i.e., the surfaces of the adapter cavity **105**) can have corresponding internal threads to secure the nut **110** within the cavity of the housing. Alternatively, these components may be unthreaded and be bolted, clamped or otherwise connected to one another. As seen in FIGS. **13** to **16**, the nut **110** includes a lip **116** extending over a perimeter rim **109** of the open end of the adapter cavity **105** to further secure the nut **110** in engagement with the coupling adapter **24**.

(62) In some embodiments, the hydraulic pressure within the adapter cavity **105** pushes the male component **14** axially toward the nut **110**. Once pressurized, the pressure inside the housing **104** is substantially uniform in all directions and pushes perpendicularly on the surfaces (e.g., against the flange portion **102**, and the radial and axial surfaces **106**, **108**). In operation, the fluid contained in the fluid passage **15** and/or fluid channel **25** is under hydraulic pressure. Operating pressures vary depending on the application, be it heavy or light. Typical ranges of hydraulic pressure in the forestry industry, for example, are between about 50 and about 4000 psi, and up to around 5000 psi in some cases. In load-sensing hydraulic circuits, the operating pressures most often vary between about 250 psi and 3000 to 4000 psi.

(63) In some embodiments, this pressure results in the direct abutment of the flange portion **102** against the projection **114** of the nut **110**. However, in this embodiment, the swivelling coupling **100** further includes a slide-ring **120** surrounding the body portion **16** and disposed between the nut **110** and the flange portion **102**. The slide-ring **120** can be adapted to reduce the friction coefficient between the components, e.g., between the flange portion **102** and the nut **110**, during rotation of

the male component **14**. The slide-ring **120** can be particularly desirable in high-pressure hydraulic systems (or alternatively in applications where the fluid is less lubricating) since the axial pressure on the male component **14** greatly increases the friction between the flange portion **102** and the nut **110**. In some embodiments, the slide-ring **120** is composed of Nylatron® material group, Nyloil™, Nycast™, Teflon, or another suitable material for such a component. In some embodiments, the slide-ring **120** can include a thrust washer adapted to support the axial load applied thereto, such as the load applied thereto by the flange portion **102** during operation and/or during hydraulic shocks. (64) Referring to FIG. **16**, when the male component **14** is coupled within the adapter cavity **105**, the flange portion **102** defines an axial play **95** with the radial surface **106** of the adapter cavity **105**. More particularly, the internal depth of the adapter cavity **105** is greater than the sum of the width of the flange portion **102** and the length of the projection **114** of the nut **110**, thus enabling an amount of axial play **95**. In some embodiments, the amount of axial play can be between about 0.005 and about 0.08 inches, although other configurations are possible, such as having a narrower axial play (e.g., <0.005 inches) or wider axial play (e.g., >0.08 inches).

(65) In this embodiment, when under internal fluid pressure, the flange portion **102** is pushed toward the projection **114** of the nut **110**, and thus abuts on the slide-ring **120**. The slide-ring **120** is adapted to enable a distribution of forces, thereby decreasing the pressure between the components. This in turn enables the swivelling coupling **100** to have improved swivelling performance at higher pressures, such as in the range of 3000 to 5000 psi, for example. In some embodiments, the slide-ring **120** has a flat disk shape, but may also have an O-ring shape to reduce the coefficient of friction. It should be noted that, when the flange portion **102** is pressed against the slide-ring **120**, the flange portion **102** is held in spaced relation to the internal axial surface **108** of the adapter cavity **105**, and corresponds to the amount of axial play **95**. It should be noted that the axial play **95** defined within the adapter cavity **105** may help protect the various components of the coupling **100** from hydraulic shocks (also called “hydraulic rams”) or other types of fluid shocks or pressure differentials in and around the coupling.

(66) In some embodiments, it can be desirable to maintain the contact between the flange portion **102**, the slide-ring **120** and the nut projection **114** to increase sealing efficiency of the coupling **100** (e.g., prevent exterior fluid(s), dust and/or debris from entering the coupling). During operation, hydraulic fluid flowing through the coupling **100** urges the flange portion **102** against the slide-ring **120**, thereby creating the contact between flange portion **102**, slide-ring **120** and nut projection **114**, and increases sealing efficiency. However, in some embodiments, the contact between these components can be broken, thereby compromising the seal integrity of the coupling. For example, and with reference to FIGS. **31** to **33**, underwater applications without any prior pressurization of the coupling (FIGS. **31** and **32**), vertical installations of the coupling where the flange portion **102** tends to move away from the slide-ring **120** under gravity (FIG. **33**), or under the effect of hydraulic shocks and a vacuum/suction effect occurs within the coupling can decrease the seal integrity of the coupling by applying pressure on inner components of the coupling and breaking the contact between the flange portion **102** and the slide-ring **120**, for example. Under such conditions, the risk of infiltration of dust, debris or other contaminants within the coupling is increased and can cause some issues.

(67) Still with reference to FIG. **16**, the swivelling coupling **100** can include a seal assembly **130** comprising at least one sealing ring **132** (or O-ring) adapted to cooperate with one or more of the male component **14**, the nut **110** and the coupling adapter **24**. In this embodiment, once the swivelling coupling **100** is assembled and in operation, the seal assembly **130** is pressed in between the male component **14** and the nut **110** to at least partially cut the pressure of any fluid leaking through the interstices of the coupling **100**. The seal assembly **130** can be adapted to prevent fluids from leaking out of the coupling **100**. In some embodiments, the seal assembly **130** may include O-rings, back-up rings, etc. Also, the seal assembly **130** may be easily replaced, in the event that it loses its efficiency. Various sealing joints may be used to cut the velocity of the fluid so that if the

fluid leaks past one of the seals, it is less likely to leak past the next seal. In other words, a series of seals may be used as part of the seal assembly **130**.

(68) In some embodiments, the seal assembly **130** further includes one or more outer seals **134** adapted to prevent fluid from leaking to an exterior of the coupling **100**, or to prevent debris and dust from entering the coupling **100**. With reference to FIGS. **13** to **16**, in this embodiment, the outer seal **134** includes a wiper ring **136** surrounding the male component **14** proximate the shank portion **18**. The wiper ring **136** is also adapted to engage the nut **110** in order to create a seal between the nut **110** and the male component **14**. In some embodiments, the wiper ring **136** can have a flat-disk shape, although other configurations are possible, as will be described below.

(69) In this embodiment, the male component **14** includes an outer groove **140** extending about the shank portion **18**. The outer groove **140** can be shaped and adapted to receive a portion of the wiper ring **136** therein. As seen in FIGS. **14** and **16**, the wiper ring **136** can include a ring portion **138** adapted to engage the nut **110**, and a protruding portion **139** extending from the ring portion **138** and engaging the outer groove **140**. It is noted that both the ring portion **138** and the protruding portion **139** surround the male component **14**, i.e., extend circumferentially around the male component. In this embodiment, the outer groove **140** includes a groove wall **142** extending relatively perpendicularly relative to the shank portion **18**, and thus substantially parallel to the front surface of the nut **110**. The wiper ring **136** is therefore shaped and adapted to extend between the nut **110** and the groove wall **142**, with the ring portion **138** engaging the nut **110**, and the protruding portion **139** engaging the groove wall **142**.

(70) In some embodiments, the wiper ring **136** is made of a resilient material, such as rubber, and is adapted to protect the various components of the coupling during operation of the hydraulic lines (e.g., while fluid is flowing between the first and second fluid lines via the coupling). During operation, it is noted that the various components of the coupling **100** can be submitted to hydraulic shocks (also called “hydraulic rams”) which can cause shifts, friction, damage, or failure of these components.

(71) Some hydraulic shocks can result from a suction effect created by the pressure differential between fluid channel **25** and the fluid passage. Specifically, hydraulic fluid can be provided to the fluid channel **25** via the second fluid line, with the valve body **70** blocking fluid flow into the fluid passage **15**. Therefore, it is noted that the pressure along the fluid channel **25** and within the adapter cavity **105** increases, while the pressure along the fluid passage **15** remains low (e.g., null). As such, when the valve body **70** is displaced in the open position, the low pressure within the fluid passage can create a vacuum causing the pressure within the adapter cavity **105** to drop substantially rapidly to match the pressure along the fluid passage **15** (e.g., 0 psi), thereby creating the suction effect within the coupling **100**.

(72) The suction effect (i.e., the hydraulic shock) often creates a force against the flange portion **102** which pushes the flange portion **102** against the nut **110**. In this embodiment, the nut **110** then pushes against the wiper ring **136** provided about the male component. The resilient material of the wiper ring **136** is configured to be deformed, thus absorbing at least some of the forces created by the hydraulic shocks and reducing movement of the male component **14**, slide-ring **120** and/or nut **110**. Once the pressure stabilizes and/or is relieved, i.e., once the forces are no longer applied on the components of the coupling, the wiper ring **136** is adapted to revert back to its initial shape and configuration, thereby also moving the nut **110** and/or male component **14** back in position.

(73) As such, it is appreciated that each time the coupling **100** is connected and/or disconnected from the first fluid line, the pressure within the adapted cavity **105** increases and decreases accordingly, thereby creating repetitive suction effects. The wiper ring **136** is configured to at least partially counteract this suction effect to protect the components of the coupling **100** (e.g., the male component **14**) and of the component(s) connected thereto (e.g., the female component), thereby increasing their lifespan and efficiency.

(74) In some embodiments, the wiper ring **136** can have a tapered shape, where the diameter of the

wiper ring at a first end thereof is greater than at a second and opposite end thereof. For example, in this embodiment, the protruding portion **139** extends at an angle from the ring portion **138** such that the inner diameter of the wiper ring **136** along the ring portion **138** is greater than the inner diameter of the wiper ring along the protruding portion **139**. Moreover, the thickness of the wiper ring **136** along the protruding portion **139** is illustratively less than the thickness thereof along the ring portion **138**. This configuration can improve the shock-absorption capabilities of the wiper ring **136** and also improve its resiliency (i.e., its ability to revert back to its initial shape and configuration). During operation, hydraulic pressure within the coupling can cause the wiper ring **136** to be squeezed between the nut **110** and the groove wall **142**, thereby causing the protruding portion **139** to at least partially deform (e.g., buckle) to absorb at least some of the forces from the hydraulic rams, for example. The protruding portion **139** is adapted to revert back to its initial configuration once pressure within the coupling has stabilized.

(75) In some embodiments, the ring portion **138** and protruding portion **139** can be made of the same material, and thereby have substantially the same resiliency and/or flexibility. However, it is appreciated that other configurations are possible. For example, the protruding portion **139** can be made of a more resilient or flexible material than the ring portion **138**. In such embodiments, it is noted that the ring portion **138** can absorb some of the forces from the hydraulic shocks, although the more resilient protruding portion **139** is configured to absorb more forces.

(76) As seen in FIG. **16**, the second spring **72** illustratively extends between an internal surface of the adapter cavity **105** and the flange portion **102**, and is adapted to also absorb some of the forces of the hydraulic shocks and bias the flange portion **102** against the slide-ring **120**. Therefore, the components of the coupling (e.g., the flange portion **102**, the slide-ring **120** and the nut **110**) are adapted to remain in contact with one another to ensure sealing efficiency of the coupling. The second spring **72** can be optional, with the axial movement of the components of the coupling being at least partially blocked by the cooperation of the wiper ring **136** with the male component **14** (e.g., with the shank portion) and the nut **110**. More specifically, under certain conditions, the flange portion **102** can be made to move into the axial play **95**, thereby increasing the risks of impacts with the housing and breaking contact with the slide-ring **120**. In this embodiment, when the male component **14** is pushed toward the axial play **95** (e.g., caused by hydraulic shocks, a vertical installation or under external pressure), the groove wall **42** engages the wiper ring **136**. The wiper ring **136** can be adapted to absorb some of the forces, and also engages the nut **110**, which in turn abuts against the housing **104** via the lip **116**. As such, axial movement of the male component **14** toward the axial play **95** is blocked, or at least reduced.

(77) It should be noted that reducing or blocking movement of the components of the coupling (e.g., the flange portion **102**, the slide-ring **120** and the nut **110**) can reduce the stress, deformation and impacts they sustain during operation. Reducing the forces applied on these components can increase their lifespan, which reduces the need to replace and/or repair them, thereby increasing the efficiency and lifespan of the coupling **100**.

(78) Referring broadly to FIGS. **1** to **16**, it should be noted that, when the male component **14** of the coupling (**10** or **100**) engages the female component of the first fluid line, the restraining mechanism **40** is adapted to prevent axial movement of the male component **14** relative to the female component (e.g., via engagement of the ball bearing assembly within the groove **42** provided about the shank portion **18**). Furthermore, the restraining mechanism **40** is also adapted to prevent rotational movement of the male component **14** about the longitudinal axis of the shank portion **18** relative to the female component via engagement of the detent **44** with the ball bearing assembly.

(79) In addition, the flange portion **102** is shaped and sized to engage the adapter cavity **105**, with the nut **110** blocking axial movement of the male component **14** toward the open end of the adapter cavity **105**, and in some embodiments, the wiper ring **136** is adapted to block, or at least impede axial movement of the male component **14** toward the coupling adapter **24** (i.e., into the axial play

95). The axial movement of the male component due to the deformation and wear of the slide-ring **120** (e.g., the thrust washer), along with the suction effect and/or atmospheric pressure (e.g., when submerged in water) is thus also at least partially managed via the wiper ring **136**, and the contact between the flange portion **102**, slide-ring **120** and nut projection **114** is maintained. It should also be noted that greater forces applied on the wiper ring **136** increase the sealing efficiency of the wiper ring **136** since the wiper ring **136** is squeezed in place between the groove wall **142** and the nut projection **114**, thereby further sealing interstices between the nut **110** and the male component **14**. This can be desirable in certain situations, such as in vertical installations, or for underwater applications, where the coupling would not be required to be pressurized prior to being submerged since the increasing pressure from the water can increase the sealing efficiency of the wiper ring **136**.

(80) Now referring to FIGS. **17** to **20A**, another implementation of the coupling **100** is shown. The male component **14** is provided within the housing **104** of the coupling adapter **24**, with the shank portion **18** extending out of the housing **104** for connecting with a female component. The coupling **100** also includes the nut **110** for connecting the male component **14** to the coupling adapter housing **104**. Referring more specifically to FIG. **20**, the male component **14** can include a plurality of parts coupled to one another. For example, in this embodiment, the male component **14** comprises an inner part **180** which includes the flange portion **102** provided in the adapter cavity **105** proximate the coupling adapter **24**, and an inner shaft **184** extending from the flange portion **102** toward the open end **107**. Additionally, the male component **14** includes an outer part **182** which includes the shank portion **18**. In this embodiment, the shank portion **18** is removably coupled to the inner part **180** and extends out the open end **107** of the adapter cavity **105**. The shank portion **18** can include an outer shaft **186** shaped and adapted to connect with the inner shaft **184**, thereby connecting the inner and outer parts of the male component **14** together.

(81) In this embodiment, the inner and outer parts **180**, **182** are connected together via an interference fit between the inner and outer shafts **184**, **186**, although other connection methods are possible, such as with the use of fasteners, via a keyjoint connection, via adhesive, etc. The inner part **180** is adapted to extend within the outer part **182** (i.e., the inner shaft **184** extends within the outer shaft **186**). However, it is appreciated that other configurations are possible, such as having the outer part **182** extend within the inner part **180** in order to connect the two parts to one another. In the illustrated embodiment, the outer shaft **186** is further adapted to engage the nut **110** such that the outer shaft **186** is coupled between the inner shaft **184** and the nut projection **114**, with the nut projection **114** being coupled between the outer shaft **186** and the housing **104**. As such, it is noted that the inner and outer parts **180**, **182** of the male component **14** are secured to the housing via their cooperation with each other and with the nut **110**.

(82) Still referring to FIG. **20**, the shank portion **18** can include a cylindrical body **188** provided between the outer shaft **186** and the port **22**. The cylindrical body **188** is configured to engage the nut **110** to further connect the outer part **182** (e.g., the shank portion **18**) to the nut **110**, and thus to the housing **104**. The cylindrical body **188** generally has a greater diameter than the outer shaft **186** and the port **22**. The nut **110** can thus have a complementary shape to the outer part **182** of the male component **14**, for example, with the projection **114** of the nut **110** having an inner diameter adapted to receive the outer shaft **186**, but prevent insertion of the cylindrical body **188** therein. The nut **110** illustratively includes an outer end **111** having a greater diameter than the projection **114** and adapted to receive therein the cylindrical body **188** of the shank portion **18**.

(83) In this embodiment, the outer end **111** defines a nut radial surface **113** adapted to face the cylindrical body **188**. As will be described further below, the male component **14** is adapted to engage the nut **110** and cooperate therewith to define the axial play **95** between the flange portion **102** and the coupling adapter housing **104**. Furthermore, the male component **14** engages the nut **110** in a manner such that the cylindrical body **188** is spaced from the nut radial surface **113**, defining therebetween a second axial play **195**. The axial play **95** and the second axial play **195** can

generally have the same dimensions, although it is appreciated that one can be greater than the other, for example.

(84) In some embodiments, the seal assembly includes various sealing elements, such as O-rings provided between the male component **14** and the nut **110**, between the nut **110** and the housing **104**, or a combination thereof. In this embodiment, and with reference to FIGS. **20** and **20A**, the coupling **100** includes a plurality of sealing rings **132**, including a pair of sealing rings **132** provided within the outer end **111** of the nut **110**, between the nut **110** and the cylindrical body **188** of the shank portion **18**. The sealing rings **132** are configured to seal interstices between the nut **110** and the male component **14** (e.g., the cylindrical body **188**), and radially restrain the male component **14** within the nut **110**. Referring more specifically to FIG. **20A**, the sealing rings **132** include a first sealing ring **132a** and a second sealing ring **132b** provided in a circumferential pocket **189** defined about the cylindrical body **188**. As mentioned, each sealing ring **132** extends between and engages the nut **110** (on an inner surface thereof) and the cylindrical body **188** (on an outer surface thereof), thereby radially restraining the cylindrical body **188** relative to the nut **110**.

(85) The first sealing ring **132a** engages substantially flat surfaces, such that it radially restrains the male component **14** within the nut **110**, along with defining a seal therebetween. In this embodiment, the outer end **111** of the nut can have a tapered inner edge **115**, with the second sealing ring **132b** engaging the tapered inner edge **115**. As such, the second sealing ring **132b** can be adapted to radially restrain the male component **14** (e.g., the outer part **182**), and at least partially provide axial restraint to the male component as well. For example, and as described above, during operation, the coupling **100** can be subject to a suction effect which can pull the male component **14** further into the adapter cavity **105**, thus pulling the cylindrical body **188** further into the outer end **111** of the nut **110**. The tapered outer edge **115** can be shaped and configured to provide an abutment for the second sealing ring **132b**, which can then block axial movement of the cylindrical body **188** toward the nut **110**. In other words, the second sealing ring **132b** is adapted to be squeezed between the lateral surface of the circumferential pocket **189** and the tapered inner edge **115**, thereby preventing axial movement of the cylindrical body **188**. It should be noted that, by blocking axial movement of the cylindrical body **188**, the outer part **182** does not push against the inner part **180**, thereby maintaining the flange portion **102** in contact with the slide-ring **120** within the adapter cavity **105**. In some embodiments, the second sealing ring **132b** can be a pressure-energized seal, at least partially blocking both axial and radial movement of the male component, and enabling greater sealing efficiency as the pressure thereon increases.

(86) It is appreciated that adjusting the angle at which the tapered inner edge **115** extends can correspondingly adjust the amount of axial restraint the second sealing ring **132b** can provide. For example, a tapered edge of about 45 degrees can be adapted to prevent axial movement over a greater range of pressures than a tapered edge of about 10 degrees. In some embodiments, the tapered inner edge **115** of the nut **110** can be angled relative to a nut axial surface **117** by about 5 to about 85 degrees, although other configurations are possible. It should also be noted that greater forces applied on the second sealing ring **132b** increase the sealing efficiency of the sealing ring **132b** since the cylindrical body **188** presses and squeezes the sealing ring **132** between the outer end of the nut **111**.

(87) This can be desirable in certain situations, such as in vertical installations (seen in FIG. **33**), or for underwater applications (seen in FIGS. **31** and **32**), where the coupling is not required to be pressurized prior to being submerged since the increasing pressure from the water increases the sealing efficiency of the second sealing ring **132b**. For example, and as seen in FIGS. **31** and **32**, one or more couplings **100** can be connected to an underwater pump, such as a pump configured to collect fish as part of a commercial fishing vessel, prior to being submerged. The pump can then be submerged without pre-pressurizing the coupling to prevent water from infiltrating the coupling. As seen in FIG. **32**, the pump can be lowered to about 20 m under the surface, prior to providing fluid through the coupling, and the sealing ring **132** (i.e., the second sealing ring **132b**) is configured to

prevent water and debris from entering the coupling **100**. It should be understood that increasing the sealing efficiency of the coupling when underwater is desirable to prevent water from infiltrating the coupling and mixing with the hydraulic fluid, which can cause issues such as freezing the fluids within the coupling, for example. It should be noted that the coupling connected to the pump can be a swivelling coupling **100**, or alternatively a non-swivelling coupling, with one or more swivelling couplings mounted along the pump conduits, such as at regular intervals (e.g., every 10 m).

(88) In some embodiments, the outer part **182** of the male component **14** is removably connected to the inner part **180** and within the nut **110**. As such, the outer part **182** can be disconnected from the inner part **180**, thereby enabling interchangeability with another part. For example, the outer part **182** can be replaced with another male component outer part **182** (e.g., for maintenance or repair), or can be replaced with a corresponding female part. With reference to FIGS. **21** and **22**, the coupling **100** can be adapted to define a female connection via the replacement of the outer part **182** of the male component, with a female component **190**. The female component **190** connects to the inner part **180** in a similar fashion as the outer part **182** described above (e.g., with an outer shaft **186** coupled between the inner shaft **184** and the nut **110**), and also engages the nut **110** via the first and second sealing rings **132a**, **132b** configured to at least partially block axial and radial movement of the female component relative to the nut **110** and housing **104**.

(89) It should be noted that the coupling can include any one of the above-described mechanisms adapted to prevent or block some form of movement of the components of the coupling. In some embodiments, the coupling can include both the restraining mechanism **40** and the outer seal **134** (e.g., the wiper ring **136**, as seen in FIG. **13**, or the tapered inner edge **115** and sealing rings **132**, as seen in FIG. **20A**), while in other embodiments, the coupling includes one of the restraining mechanism **40** and the outer seal **134**, as seen in FIG. **1** (i.e., restraining mechanism **40** only), for example. It should also be noted that the coupling can be adapted to provide connection between a pair of fluid lines, as described herein, although other configurations are possible. For example, the coupling may also interconnect more than two fluid lines, where the male component **14** can include a plurality of ports **22** and can be connectable to a corresponding number of fluid lines. In some embodiments, the coupling can be coupled between an apparatus and a first hydraulic line, whereby the apparatus provides the flow of hydraulic fluid.

(90) Referring to FIGS. **23** to **25C**, the quick release male coupling and related components (e.g., the wiper ring **136** and/or the tapered inner edge **115** and sealing rings **132**) and mechanisms (e.g., the restraining mechanism **40**) described herein can be included in various types of couplings adapted for various applications. For example, the restraining mechanism **40** can be provided on a coupling cartridge **10a** (FIG. **23**) adapted for connection with a coupling housing, such as a swivelling coupling housing (FIG. **24**) adapted for applications requiring manipulations of the hydraulic lines in a 3D space, such as in the forest industry. In other embodiments, the coupling can include a manifold-type coupling, which can be a quick release male coupling (FIG. **25**).

Alternatively, and with reference to FIGS. **25B** and **25C**, the coupling can correspond to a swivelling manifold-type coupling which can include quick-lock (or quick-release) capabilities **100b**, threaded male connection(s) **100c**, a female connection **100d** and/or a «screw-to-connect coupling with swivel» **100e**. However, it is appreciated that other configurations are possible, and that the couplings can have any suitable size and/or shape. In addition, and as seen in FIG. **26**, the coupling can be part of a multiport assembly **150**, whereby a plurality of couplings, such as swivelling couplings **100**, can be coupled to a locking plate **152** which can be connected to hydraulic machinery in any suitable manner.

(91) Now referring to FIGS. **27** to **30**, the coupling **10** can be provided with a crimp fitting **160** attachable to one end of the coupling **10**, such as to the male component **14**. It should be noted that the crimp fitting, or crimped sleeve **160**, is adapted to create custom hydraulic hose assemblies, and defines a crimped end **162** of the coupling **10** (i.e., the end provided with the crimp fitting **160**)

adapted to be connected to a hydraulic line (e.g., the first fluid line). The crimped end **162** is securable to the hydraulic line via a crimping machine or device, creating a permanent connection therebetween for repairing or creating hydraulic line assemblies. In this embodiment, the coupling can include a plurality of sealing rings **132**, including an intermediary sealing ring **132c** (e.g., an O-ring) between the crimped sleeve and the projection **114** of the nut **110** configured to block axial movement of the crimped sleeve **160** relative to the nut **110** and housing **104**, thereby blocking axial movement of the male component **14** in a similar fashion.

(92) It is appreciated that in the implementations described herein, the couplings can have increased sealing efficiency against external pressures, such as in underwater applications, for example. Moreover, the sealing efficiency of these couplings can increase as the surrounding pressure increases as the seals surrounding the coupling are squeezed into position, further sealing the interstices in the coupling. In some embodiments, the sealing elements can provide axial and radial contact with one or more component of the coupling. These sealing elements can be adapted to at least partially block radial and axial movement of the one or more components, thereby reducing the stress applied thereon and increasing their lifespan.

(93) The present disclosure may be embodied in other specific forms. The described example implementations are to be considered in all respects as being only illustrative and not restrictive. For example, it should be understood that the shape of the male component **14** may be modified in a plurality of ways. Similarly, the internal shape of the adapter cavity **105** can be modified and should have a corresponding shape to house the flange portion **102**, and to provide support surfaces for distributing forces while allowing adequate play for lubrication and movement. Furthermore, the flange portion **102** may have surfaces that are bevelled or curved. These bevelled or curved surfaces can affect the fluid movement between the surfaces of the coupling and also affect the force distribution. Since the pressurized fluid exerts pressure perpendicularly with respect to the solid surfaces of the coupling components, it is appreciated that varying the angles and curvatures and/or surface area of the male component, different force-distribution effects may be achieved.

(94) In addition, one or more of the components of the coupling(s) and/or steps of the method(s) described herein could be modified, simplified, altered, omitted and/or interchanged, without departing from the scope of the present disclosure, depending on the particular applications which the coupling is intended for, and/or the desired end results, as briefly exemplified herein and as also apparent to a person skilled in the art. For example, the restraining mechanism **40** can be modified to enable interaction and cooperation with a different component than a ball bearing assembly. As seen in FIGS. **34** and **35**, in some embodiments, the internal components of the female portion can include one or more latching elements **300** having an elongate shape, and which can extend at least partially circumferentially within the port of the female portion, for example. As such, the male component can include complementarily-shaped detents (e.g., recesses **52**) for receiving therein the latching elements **300**. It is noted that the latching elements **300** can be provided at regular intervals about the male component **14** (e.g., about the shank portion **18**), such as at 90 degrees intervals, although other configurations are possible.

(95) The present disclosure intends to cover and embrace all suitable changes in technology. The scope of the present disclosure is, therefore, described by the appended claims rather than by the foregoing description. The scope of the claims should not be limited by the implementations set forth in the examples, but should be given the broadest interpretation consistent with the description as a whole.

(96) As used herein, the terms “coupled”, “coupling”, “attached”, “connected” or variants thereof as used herein can have several different meanings depending in the context in which these terms are used. For example, the terms coupled, coupling, connected or attached can have a mechanical connotation. For example, as used herein, the terms coupled, coupling or attached can indicate that two elements or devices are directly connected to one another or connected to one another through one or more intermediate elements or devices via a mechanical element depending on the particular

context.

(97) In the above description, the same numerical references refer to similar elements. Furthermore, for the sake of simplicity and clarity, namely so as to not unduly burden the figures with several references numbers, not all figures contain references to all the components and features, and references to some components and features may be found in only one figure, and components and features of the present disclosure which are illustrated in other figures can be easily inferred therefrom. The implementations, geometrical configurations, materials mentioned and/or dimensions shown in the figures are optional, and are given for exemplification purposes only.

(98) In addition, although the optional configurations as illustrated in the accompanying drawings comprises various components and although the optional configurations of the couplings as shown may consist of certain geometrical configurations as explained and illustrated herein, not all of these components and geometries are essential and thus should not be taken in their restrictive sense, i.e. should not be taken as to limit the scope of the present disclosure. It is to be understood that other suitable components and cooperations thereinbetween, as well as other suitable geometrical configurations may be used for the implementation and use of the couplings, and corresponding parts, as briefly explained and as can be easily inferred herefrom, without departing from the scope of the disclosure.

Claims

1. A quick release coupling for providing a connection between a first fluid line provided with a female component comprising a ball bearing assembly and a second fluid line, the quick release coupling comprising: a male component having a fluid passage therethrough and comprising: a body portion having an internal surface at least partially defining the fluid passage; and a shank portion extending from the body portion and being adapted to engage the female component of the first fluid line, the shank portion having a shank portion port adapted to establish fluid communication between the first fluid line and the fluid passage, the shank portion comprising a restraining mechanism adapted to engage the ball bearing assembly and block a rotational movement and an axial movement of the male component relative to the first fluid line when engaging the first fluid line; and a coupling adapter having a fluid channel therethrough and being securable to the male component proximate the body portion to establish fluid communication between the fluid channel and the fluid passage, the coupling adapter being adapted to engage the second fluid line and comprising an adapter port adapted to establish fluid communication between the second fluid line and the fluid channel, the coupling adapter further comprising a housing having internal surfaces comprising at least one radial surface and at least one axial surface, said internal surfaces defining a cavity having an open end, wherein the body portion of the male component is shaped and sized to engage the cavity via the open end and abut the at least one radial surface; and a nut securable within the cavity of the housing and surrounding a section of the shank portion to radially restrain the male component, the nut being adapted to axially restrain the body portion within the cavity and allow the male component to rotate with respect to the housing and the nut about a longitudinal axis of the shank portion, thereby enabling the first and second fluid lines to be swivellably interconnected.

2. The quick release coupling according to claim 1, wherein the shank portion comprises an external surface, and wherein the restraining mechanism is provided along the external surface of the shank portion.

3. The quick release coupling according to claim 1, wherein the restraining mechanism comprises a groove extending circumferentially about the shank portion, the groove being shaped and adapted to receive the ball bearing assembly for blocking the axial movement of the male component.

4. The quick release coupling according to claim 3, wherein the groove extends circumferentially about the shank portion in a single plane.

5. The quick release coupling according to claim 3, wherein the groove extends partially about the shank portion such that the groove comprises a first end and a second end spaced from one another and defining an ungrooved portion therebetween.
6. The quick release coupling according to claim 5, wherein the ungrooved portion is shaped and adapted to engage the ball bearing assembly between a pair of adjacent bearing balls of the ball bearing assembly to block the rotational movement of the male component.
7. The quick release coupling according to claim 1, wherein the restraining mechanism further comprises a detent adapted to engage the ball bearing assembly to block the rotational movement of the male component.
8. The quick release coupling according to claim 7, wherein the detent comprises a plurality of recesses provided along a groove for receiving respective bearing balls of the ball bearing assembly to block the rotational movement of the male component.
9. The quick release coupling according to claim 7, wherein the detent comprises a protrusion extending from the shank portion, the protrusion being shaped and sized to engage the ball bearing assembly between a pair of adjacent bearing balls of the ball bearing assembly to block the rotational movement of the male component.
10. The quick release coupling according to claim 9, wherein the restraining mechanism comprises a groove extending circumferentially about the shank portion, and wherein the protrusion extends from within the groove.
11. The quick release coupling according to claim 9, wherein the protrusion is integrally formed with the shank portion or is welded thereto.
12. The quick release coupling according to claim 1, further comprising a valve assembly operable to control fluid flow along the fluid passage of the male component before, during and after engaging the shank portion with the first fluid line.
13. The quick release coupling according to claim 12, wherein the valve assembly comprises a valve head provided within the fluid passage proximate the shank portion port and being operable in a closed position where fluid flow is blocked through the shank portion port, and an open position, the valve assembly further comprising a head spring connected to the valve head and being adapted to bias the valve head in the closed position.
14. The quick release coupling according to claim 12, wherein the valve assembly comprises a valve body provided within the fluid passage along the body portion and being operable in a closed position where fluid flow through the fluid passage is blocked, and an open position where fluid flow through the fluid passage is allowed, the valve assembly further comprising a body spring connected to the valve body and being adapted to bias the valve body in the closed position.
15. The quick release coupling according to claim 1, further comprising a thrust washer surrounding the shank portion between the nut and the body portion.
16. The quick release coupling according to claim 1, wherein the shank portion comprises a flange extending radially outward and having a flange surface being substantially perpendicular to an external surface of the shank portion and facing the nut, and wherein the quick release coupling further comprises an outer sealing element surrounding the shank portion and extending between the flange surface and the nut.
17. A quick release coupling for providing a connection between a pair of hydraulic lines, the quick release coupling comprising: a male component having a fluid passage therethrough and comprising: a shank portion adapted to engage a first hydraulic line, the shank portion having a first port adapted to establish fluid communication between the first hydraulic line of the pair of hydraulic lines and the fluid passage, the shank portion comprising a restraining mechanism adapted to cooperatively engage the first hydraulic line and block a rotational movement and an axial movement of the male component relative to the first hydraulic line when engaged with the first hydraulic line; and a coupling adapter connectable to the male component and having a fluid channel therethrough adapted to be in fluid communication with the fluid passage, the coupling

adapter being adapted to engage a second hydraulic line of the pair of hydraulic lines and comprising a second port adapted to establish fluid communication between the second hydraulic line and the fluid channel, the coupling adapter further comprising a housing having internal surfaces defining a cavity having an open end, wherein the male component is shaped and sized to engage the cavity via the open end and abut at least one of the internal surfaces; and a nut securable within the cavity of the housing and surrounding a section of the shank portion to radially restrain the male component, the nut being adapted to axially restrain the male component within the cavity and allow the male component to rotate with respect to the housing and the nut about a longitudinal axis of the shank portion, thereby enabling the first and second fluid lines to be swivellably interconnected.

18. A quick release coupling for providing a connection between a pair of hydraulic lines, the quick release coupling comprising: a male component having a fluid passage therethrough and comprising: a shank portion adapted to engage a first hydraulic line of the pair of hydraulic lines, the shank portion having a first port adapted to establish fluid communication between the first hydraulic line and the fluid passage, the shank portion comprising a restraining mechanism adapted to cooperatively engage the first hydraulic line and block a rotational movement and an axial movement of the male component relative to the first hydraulic line when engaged with the first hydraulic line; and a coupling adapter comprising a housing shaped and adapted to receive the male component therein and having a fluid channel therethrough adapted to be in fluid communication with the fluid passage, the coupling adapter being configured to axially restrain the male component while enabling rotational movement of the male component within a cavity and about a longitudinal axis of the shank portion, the coupling adapter being adapted to engage a second hydraulic line of the pair of hydraulic lines such that the first and second hydraulic lines are swivellably connected, the coupling adapter further comprising a second port adapted to establish fluid communication between the second hydraulic line and the fluid channel.
